IF2240 Basdat

Introduction to Multidimensional Data Model & Data Warehouse

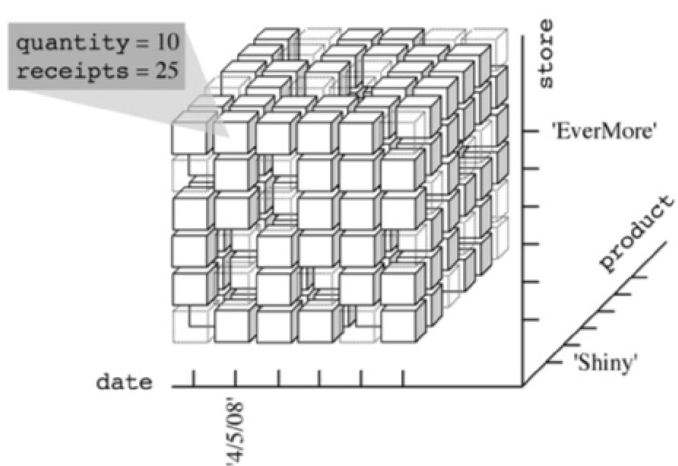
Multidimensional

- The multidimensional model begins with the observation that the factors affecting decision-making processes are enterprise-specific facts, such as sales, shipments, hospital admissions, surgeries, dan so on. Instances of a fact correspond to events that occurred.
- For example, every single sale or shipment carried out is an event. Each fact is described by the values of a set of relevant measures that provide a quantitative description of events. For example, sales receipts, amounts shipped, hospital admission costs, and surgery time are measures.
- Used metaphor of **cubes to represent multidimensional data**. Events are associated with **cube cells**. Each cube cell is given a value for each measure. **Cube edges** stand for analysis dimensions. If more than three dimensions exist, the cube is called a *hypercube*.

Terminology

- Dimension: subject label for a row or column
- Member: value of dimension
- Measure: quantitative variables stored in cells

Sales Data Cube Example



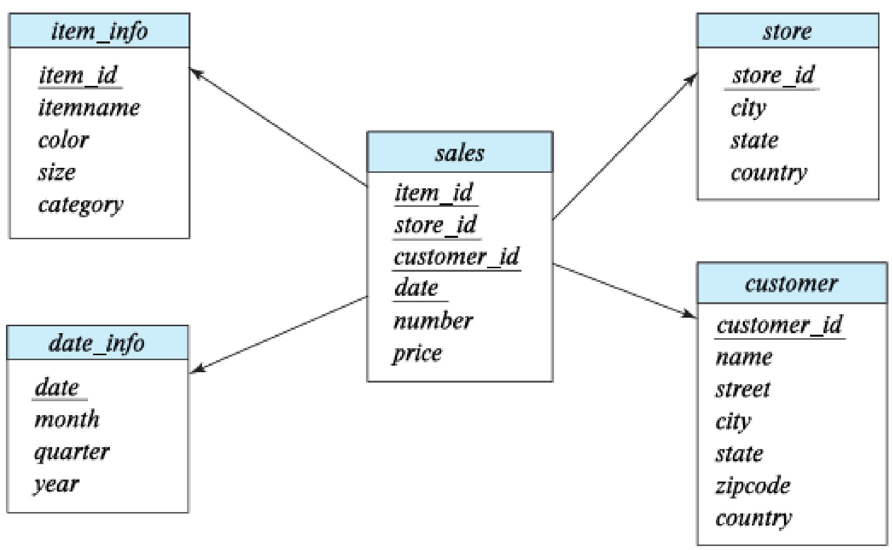
Relation schema: SALES(<u>store</u>, <u>product</u>, <u>date</u>, quantity, receipts)(i.e. <'EverMore', 'Shiny', '04/05/08', 10, 25>)

Multidimensional Data and Warehouse Schemas

Data in warehouse can usually be divided into:

- **Fact tables**, which are large. e.g. SALES(item_id, store_id, customer_id, date, number, price)

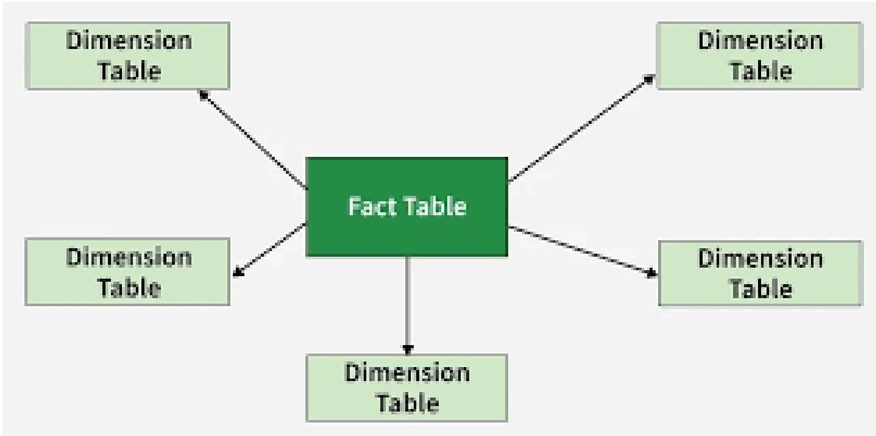
  Attributes of fact tables can be usually viewed as:
  - **Measure attributes**
    - measure some value and can be aggregated,
    - e.g., the attributes number or price of the sales relation
  - **Dimension attributes**
    - dimensions on which measure attributes are viewed
    - e.g., attributes item_id, color, and size of the sales relation
    - usually small ids that are foreign keys to dimension tables.
- **Dimension tables**, which are relatively small (e.g. store extra information about stores, items, etc.)
- The warehouse schema for previous sales data cube is as below:



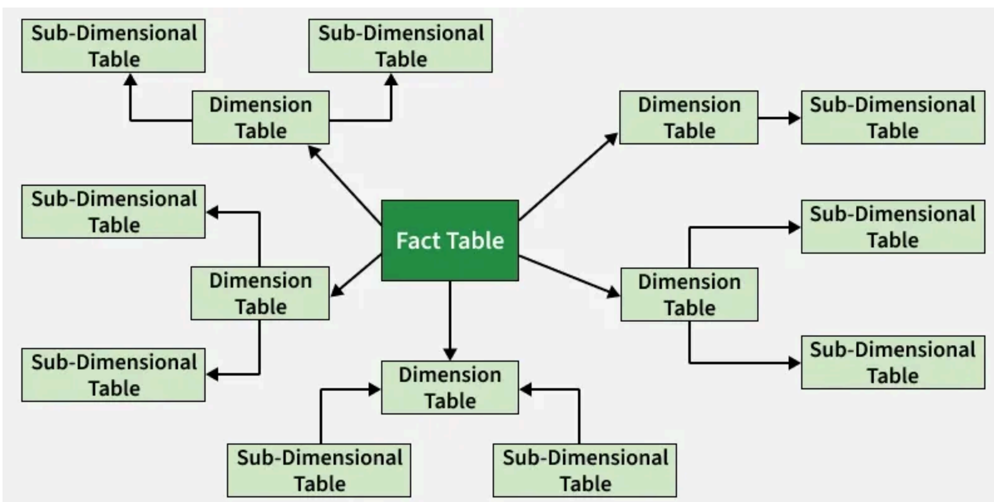
Conceptual Modeling of Data Warehouses

Modeling data warehouses: dimensions & measures

- Star schema

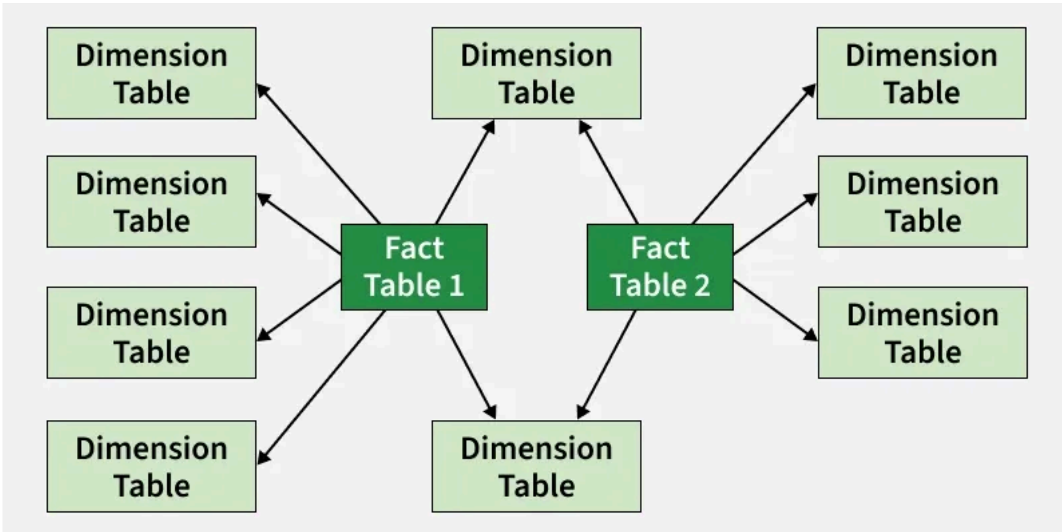


  - A fact table in the middle connected to a set of dimension tables
- Snowflake schema



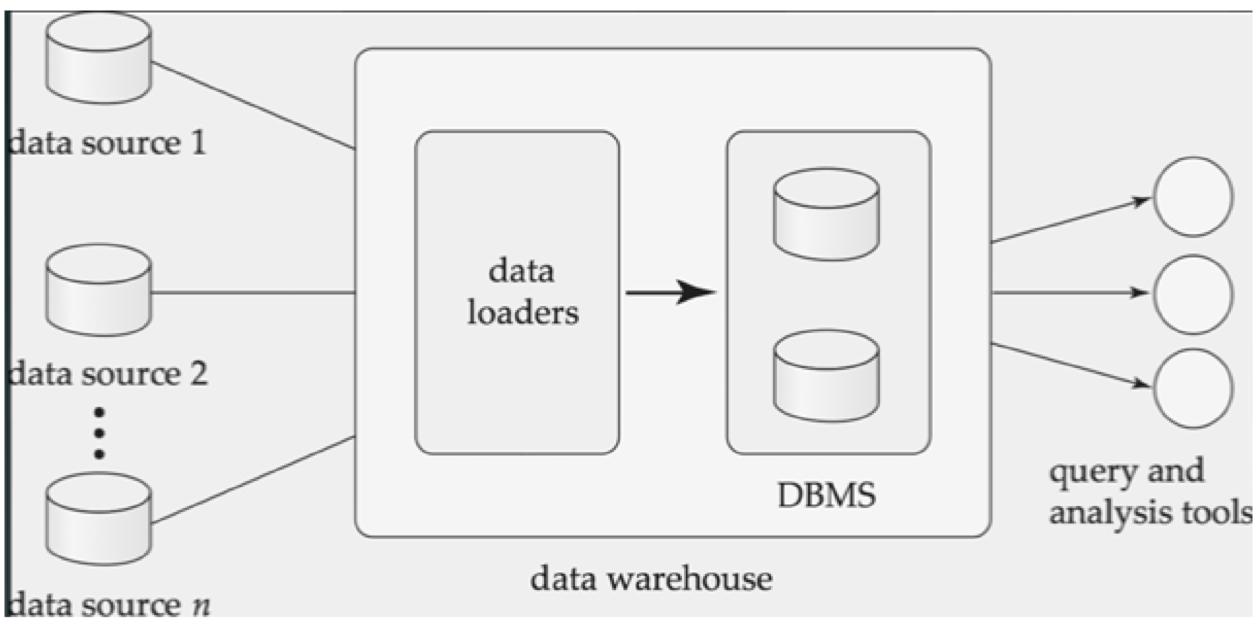
  - A refinement of star schema where some dimensional hierarchy is normalized into a set of smaller dimension tables,

- Fact constellations



  - Multiple fact tables share dimension tables, viewed as a collection of stars, therefore called galaxy schema or fact constellation

Data Warehousing



Data Warehouse vs. Operational DBMS

OLTP vs. OLAP

	OLTP	**OLAP**
Users	Clerk, IT professional	Knowledge worker
Function	Day to day operations	Decision support
DB Design	Application-oriented	Subject-oriented
Data	Current, up-to-date Detailed, flat relational Isolated	Historical Summarized, multidimensional Integrated, consolidated
Usage	Repetitive	Ad-hoc
Access	Read/Write, Index/hash on primary key	Lots of scans
Unit of Work	Short, simple transactions	Complex query
# Records accessed	Tens	Millions
# users	Thousands	Hundreds
DB size	100MB-GB	100GB-TB
Metrics	Transaction throughput	Query throughput, response